

Spatial Interpolation Model (MKM-SP-001)

Wolfram Oracle Benchmark of the Code and its v1.0 Document

MKM Research Labs

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David K Kelly

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1 Executive Summary

This document benchmarks the Spatial Interpolation Model (MKM-SP-001): the module `src/models/floodrisk/spatial.py`. It asks two questions. Does the code compute what its model document (`spatial_model.pdf`, v1.0, 27-March-2026) says? And does what the document says hold?

The code computes the document's distance and IDW. The haversine (Eq. 2) and inverse-distance weighting (Eq. 1) agree with a 50-digit implementation of the document's equations to 1.5×10^{-14}. That holds both for gauges given as distances and for gauges given as coordinates. Wolfram, computing each independently, agrees with the reference to 3.2×10^{-31}. The three sensitivity tables reproduce: all 26 figures.

The document's claims do not hold: 5 of 15 are met (Table 2).

- **The spatial index finds the wrong gauge.** The KDTree indexes latitude and longitude as though a degree of each were the same length. Over the Thames basin it names the wrong nearest gauge in 176 of 1000 queries.
- **The correlation is not Eq. 3.** The code measures distance as Euclidean degrees at 111 km a degree, so east–west separations are overstated by 61% at 51.5°N. Against Eq. 3 on the document's own distance, it is up to 98% off. MKM-DD-001 v2.0 already documents the code's convention.
- **The interpolation is not configurable from coordinates.** The power and the 1 m threshold are fixed in the path that takes coordinates.
- **No function returns the nearest gauge.** §4.2 lists the nearest gauge's distance and identity as outputs.
- **The haversine is not exact.** It assumes a sphere, against Q2's statement, though it is sufficient at these scales, as §5.1 says.
- **The model is not used for its stated purpose.** Nothing in the platform calls the interpolation, the index or the correlation. What is used is the haversine and the polyline projection, by the property, gauge and seismic code.
- **The validation evidence does not exist.** No test queries the tree (Q4), nothing validates the interpolation against observations (Q5), and nothing compares it with kriging (Q7).

The projection's flat-Earth error, which the document flags beyond 50 km, is real but small.

Evidence. 7 checks, outcome: 1 fail (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `7bbb63078a59`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-11-06 by `python -m tools.wolfram_oracle sp001`.

2 Model Purpose and Scope

2.1 What is benchmarked

Every function in `spatial.py`:

- `haversine_distance`;

- the interpolation, from coordinates (`spatial_interpolate_wse`) and from distances (`idw_interpolate_from_gauges`);
- the spatial index (`build_spatial_index`);
- the correlation matrix (`build_correlation_matrix`);
- the projection onto a segment and a polyline (`nearest_point_on_segment`, `nearest_point_on_polyline`).

2.2 What is not

- Whether IDW suits flood surfaces: the document's assumptions of smooth variation and isotropy (§5) are judgements, not computations.
- Q3's gauge metadata, which belongs to the gauge pipeline.

This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

The reference implements the document's equations directly from its text:

- the haversine of Eq. 2, in its arctangent form, on a sphere of 6,371,000 m;
- the IDW of Eq. 1 with that distance and power p. The first gauge closer than the threshold (default 1 m) returns its own value;
- a gauge's water surface elevation as its depth plus its elevation (§4.1);
- the correlation of Eq. 3, $\rho_0 \exp(-d/L)$, with d the distance of Eq. 1;
- the nearest gauge by that distance (§3.3, §4.2);
- for the projection, the true nearest point on a segment drawn straight in latitude and longitude, found by minimising the haversine distance along it. That is what the flat-Earth projection approximates.

4 Benchmark Method

The cases.

- IDW: six gauge sets at seven powers from 0.5 to 4, including gauges inside the threshold and on it. The coordinate path: four gauges around a target on the Thames, with the target on a gauge, 0.5 m from one, far off, and with a single gauge.
- The correlation: five properties within 15 km, $L = 2$ km, $\rho_0 = 0.6$.
- The spatial index: one case built to expose it, a gauge 0.010° north and one 0.012° east of the target. Then 20 seeded random networks of 50 gauges over the Thames basin, queried at 50 random points each.
- The projection: four segments (east–west, north–south, diagonal and 1 km long). Points sit 10 m to 200 km from each midpoint, to the north, east and north-west. Nine points are projected onto a river-like polyline.

The printed tables. Tables 1–3 were generated from the code. Each figure is judged as the code's value, rounded and formatted as their generator did.

The statements.

- What the model returns and accepts is judged from the code.
- What uses the model is judged by searching `src` and `scripts` for callers of each function.
- Q4, Q5 and Q7 are judged by searching the repository. Q4 needs a test that queries the tree, Q5 any code that calls the interpolation, and Q7 any mention of kriging.
- "Sufficient" (§5.1) has no figure in the document. It is judged here as within 0.5% of WGS84.

Wolfram. Wolfram works from each input's exact value, independently of the reference:

- the great-circle distance, as R times the angle between the two points' unit vectors (not the haversine);
- Eq. 1 at powers 2 and 5/2;
- Eq. 3 on that distance;
- the nearest point on the diagonal segment, by `FindMinimum`;
- the true distance between six UK pairs under 100 km, by `GeoDistance` on WGS84.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Eq. 2, the haversine: code against the reference, worst error (m)	9.0×10^{-10}	1.0×10^{-6}	pass
Eq. 1, IDW from distances, seven powers: worst relative error	1.8×10^{-16}	1.0×10^{-12}	pass
§3.1, IDW from coordinates: WSE as depth plus elevation, Eq. 2's distance: worst relative error	1.5×10^{-14}	1.0×10^{-12}	pass
The correlation, against the 111 km-a-degree convention it follows: worst relative error	1.3×10^{-15}	1.0×10^{-12}	pass
Eq. 3, the correlation as the document states it: worst relative error	9.8×10^{-1}	1.0×10^{-12}	**FAIL**
The polyline projection picks the reference's segment (9 points): segments that differ	0	0	pass
Wolfram agrees with the reference	3.2×10^{-31}	1.0×10^{-25}	pass

The code computes the IDW and the haversine exactly. The correlation agrees exactly with the 111 km-a-degree convention and not with Eq. 3, so the gap is in the formula, not the arithmetic.

5.1 The model document's claims

Table 2: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
Table 1	WSE by IDW power	14 of 14 figures reproduced	met
Table 2	WSE by distance of the nearest gauge	7 of 7 figures reproduced	met
Table 2	the gauge varied is the nearest	in 5 of 7 rows; at 5,000 and 10,000 m the 2,000 m gauge is nearer	**not met**
Table 3	WSE by number of gauges	5 of 5 figures reproduced	met
§3.1, §4.1	the power and the 1 m threshold are inputs	fixed at 2 and 1 m where gauges are given by coordinates; inputs only where they are given by distance	**not met**
§3.3	the KDTree answers nearest-neighbour queries	it measures degrees of latitude and longitude alike, and names the wrong gauge in 176 of 1000 queries, up to 55% farther; in the case built for it, the gauge 1112 m away over one 831 m away	**not met**
§3.4, Eq. 3	correlation decays with the distance d of Eq. 1	the code converts Euclidean degrees at 111 km a degree; up to 98% off Eq. 3 east–west	**not met**
§4.2	it returns the nearest gauge's distance and identity	no function returns a gauge	**not met**
§5.1(2)	the haversine is sufficient below 100 km (judged: within 0.5% of WGS84)	worst -0.32%	met
Q2	the haversine is exact for the Earth's geometry	a sphere: -0.32% from WGS84	**not met**
§5.2(3)	the flat-Earth projection errs beyond 50 km	worst 0.0012% within 50 km; 0.019% (39 m) at 200 km	met
§1, §2.1, Q1	it maps gauge readings to properties by IDW, its exclusive use	nothing in `src` or `scripts` calls the interpolation, the index or the correlation; the haversine and projection serve the gauge, property, seismic code	**not met**
Q4	unit tests verify KDTree correctness	no test queries the tree	**not met**
Q5	outcomes are validated against held-out gauge readings	no code calls the interpolation to compare it with anything	**not met**
Q7	IDW is compared against kriging and nearest-neighbour baselines	kriging appears nowhere in the repository	**not met**

What the tables do not show. Table 2 calls its varied gauge the nearest. Beyond 2 km the 2 km gauge is nearer, so its last two rows measure something else. Tables 2 and 3 print to 0.1 m, which hides most of what they vary: Table 3 prints 12.4 four times over a change of 7 cm.

Empty input. With no gauges, `spatial_interpolate_wse` returns 0 m. That is a valid water surface elevation (sea level), not a missing value. The document does not mention it.

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute the document's distance and IDW. Whether IDW represents a flood surface is not tested, and nothing in the repository tests it.

2. **Searches are of this repository.** “Nothing calls it” means nothing in PhysicalRisk at the commit above.

3. **The spatial index is judged on random networks.** How often it errs depends on gauge density and latitude. The figure is for the Thames basin.

7 Document History

Date	Version	Description	Author
06-Nov-2026	0.1	First issue: tests the filed v1.0	David K Kelly